Table 12.2 Differences between Dicot and Monocot root

S. No.	Tissues	Dicot Root	Monocot Root
1	Number of Xylem	Tetrarch	Polyarch
2	Cambium	Present (During secondary growth only)	Absent
3	Secondary Growth	Present	Absent
4	Pith	Absent	Present
5	Conjunctive Tissue Ex.	Parenchyma Bean	Sclerenchyma Maize

(ii) Middle cortex: It is made up of few layers of chlorenchyma cells. It is involed in photosynthesis due to the presence of chloroplast.

(iii) Inner cortex: It is made up of few layers of parenchyma cells. It helps in gaseous exchange and stores food materials.

Endodermis is the inner most layer of cortex it consists of a single layer of barrel shaped cells, these cells contain starch grains. So it is also called **starch sheath**.

3. **Stele:** The central part of the stem inner to endodermis is known as stele. It consists of pericycle, vascular bundle and pith.

(i) Pericycle: It occurs between vascular bundle and endodermis. It is multilayered, parenchymatous with alternating patches of sclerenchyma.

(ii) Bundle Cap: There is a patch of hard sclerenchyma tissue outside to the phloem of vascular bundle is calld Bundle Cap.

(iii) Vascular bundle: Vascular bundles are conjoint, collateral, endarch and open. They are arranged in the form of a ring around the pith.

(iv) Pith: The large central parenchymatous zone with intercellular spaces is called pith. It helps in the storage of food materials.

12.6 Internal Structure of Monocot Stem (Maize)

A transverse section of monocot stem reveals the following structures.

1. **Epidermis:** It is the outermost layer. It is made up of single layer of parenchyma cells. It is covered with thick cuticle. Multicellular hairs are absent and stomata are also less in number.
2. **Hypodermis:** It is made up of few layers of sclerenchyma cells interrupted by chlorenchyma. Sclerenchyma provides mechanical support to plant.
3. **Ground tissue:** The entire mass of parenchyma cells next to hypodermis

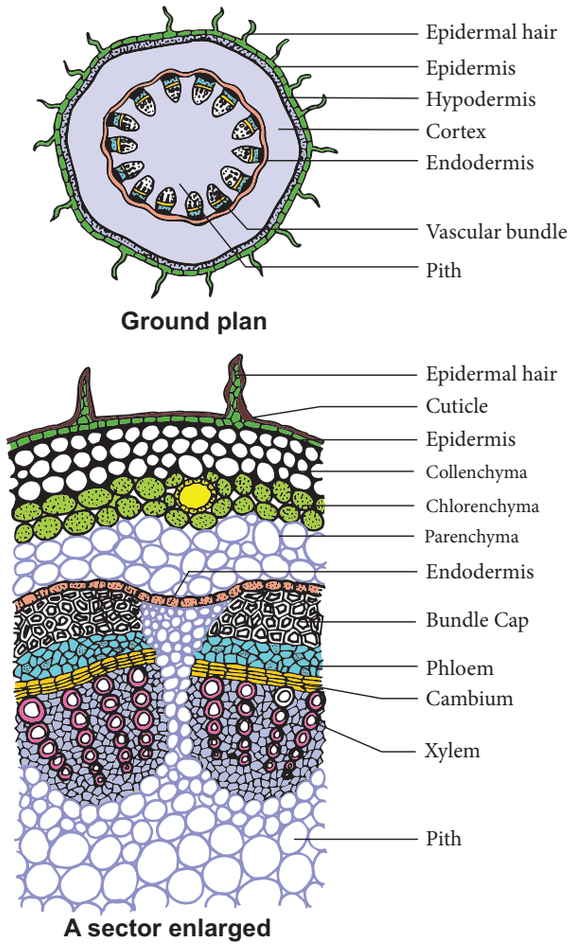


Figure 12.4 Transverse section of Dicot stem

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